UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

MAY EXAMINATION

# UCCN2243 INTERNETWORKING PRINCIPLES AND PRACTICES

THURSDAY, 11 MAY 2023 TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF COMPUTER SCIENCE (HONOURS)

**Instruction to Candidates :**

This question paper consists of THREE (3) questions in Section A and TWO (2) questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than ONE (1) question in section B, marks will only be awarded for the FIRST question in that section in the order the candidate submits the answers.

Candidates are allowed to use calculator.

Answer questions only in the answer booklet provided.

## Section A (Compulsory Questions)

Q1. You are assigned the following network 200.1.152.0/21. Using variable-length subnet masking (VLSM), create subnets as required for the network shown in Figure 1 (assume IP subnet-zero). Ensure that you always utilize the next available network in your IP plan.

(a) **Calculate** the network address, and subnet masks for all the networks (Area 1 to Area 6). (14 marks)

(b) **Determine** all the gateway addresses on the router interface G1. Assume the respective gateway addresses is the last usable address. (8 marks)

(c) **Determine** the gateway address on the router interface G2. Assume the respective gateway address is the second last usable address. (3 marks)

[Total : 25 marks]

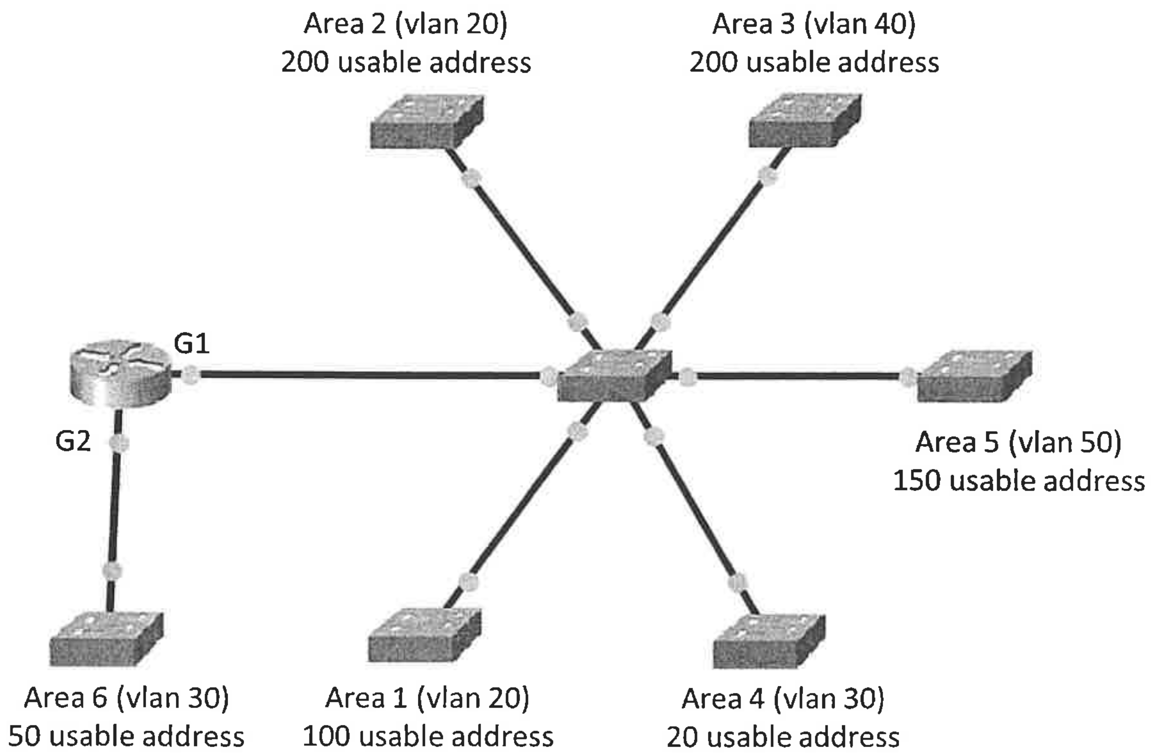


**Figure 1**

Q2. Refer to the network in Figure 2. Company AA has a public IP address of 200.1.1.1. Network Address Translation (NAT) is configured on routers R1 and R2 to enable Internet access of Company AA server and hosts using the public IP address.

(a) Due to security reasons, R1 will only translate hosts with the IP address 192.168.1.1 to the public IP for Internet access. **Show** the relevant ACL and Port Address Translation configurations on routers R1 and R2. Assume the necessary IP addresses have been configured in all the router interfaces. (Hint: You will only need to utilize the information given in Figure 2 for your setup). (15 marks)

(b) **Show** the configuration of router R1 supposed it uses a static route and a default route to connect with the local networks and the Internet. (6 marks)

(c) Continue from Q2.(b), **show** the relevant Port Forwarding configurations of router R1 to enable traffic from the Internet to access the HTTPS server. (4 marks)

[Total : 25 marks]

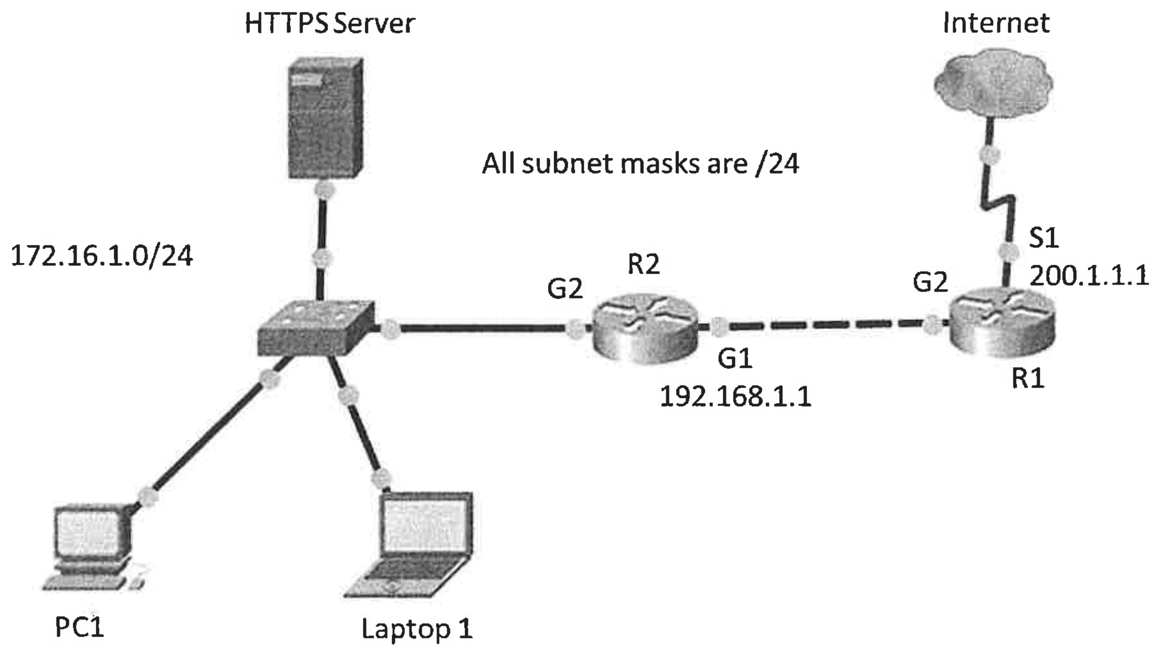


**Figure 2**

Q3. (a) An IPv6 only network is shown in Figure 3. Answer the following questions:

(i) **Determine** the IPv6 Auto Config address for PC1 and PC2 if supposed the gateway address for Subnet 1 is 2001::1000/64. (6 marks)

(ii) Let the interface address for router R1 in Subnet 2 be 2002::ACEE:1000/106. **Show** all the necessary commands to configure the remaining interfaces of routers R2 and R3 in Subnet 2 with the second and third usable IPv6 address respectively. (8 marks)

(iii) **Show** the static route configurations in R1 and R3 to enable PC1 to communicate with Server 1. (Hint: Assume the interface of router R4 in Subnet 4 is the first usable IP.) (6 marks)

(b) Briefly **describe** FIVE (5) general features of IPv6 anycast address. (5 marks)

[Total : 25 marks]

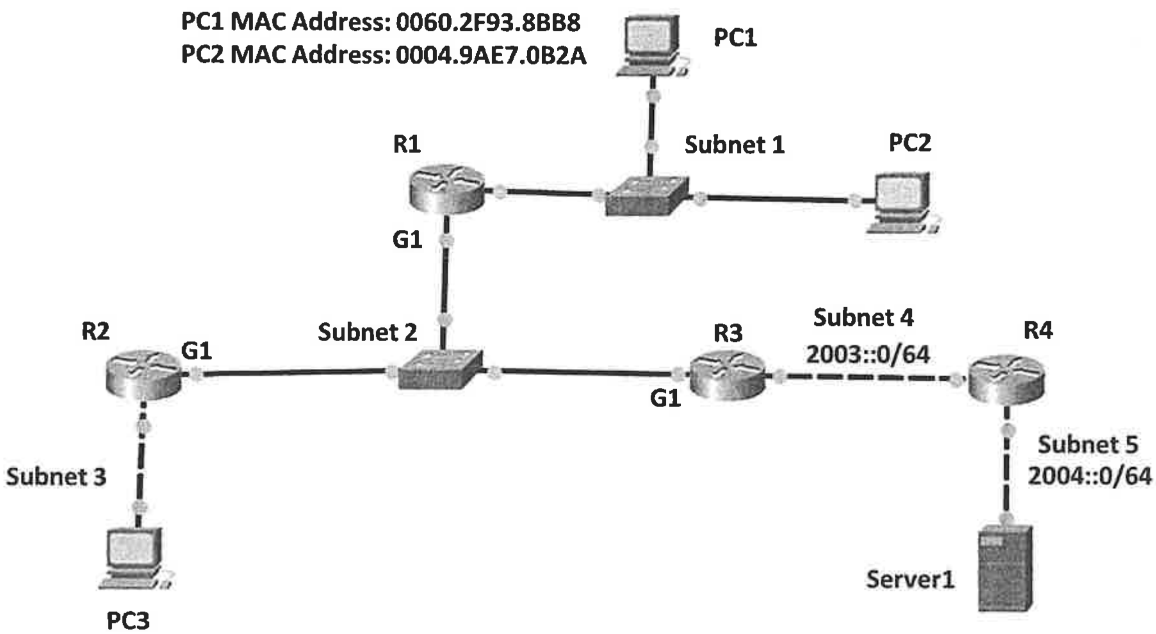


Figure 3

**Section B (Choose Any One Question)**

Q4. (a) A PC has been configured with the following iptables commands as shown in Table 1. **Explain** the actions of these commands on the specified network traffic to the PC. (6 marks)

```
#iptables –A INPUT –p icmp --icmp-type echo-request -m limit --limit 2/second –j ACCEPT

#iptables –A INPUT –p icmp --icmp-type echo-request -m limit --limit 3/minute –j LOG
```

**Table 1**

(b) Given the network shown in Figure 4.

(i) Utilizing the additional network components provided (i.e. Network Tab, Network Administrator and Network Cable), **redraw** the network to enable the Network Administrator to monitor the traffic of PC1, PC2 and PC3 using the Network Tap provided. (10 marks)

(ii) **Redraw** the network of Figure 4 again, supposing the Network Administrator would just want to monitor the traffic of only PC1, using the Network Tap provided. (6 marks)

(iii) **List** THREE (3) other methods to capture traffic from a target device on a switched network (3 marks)

Q4. (b)(Continued)

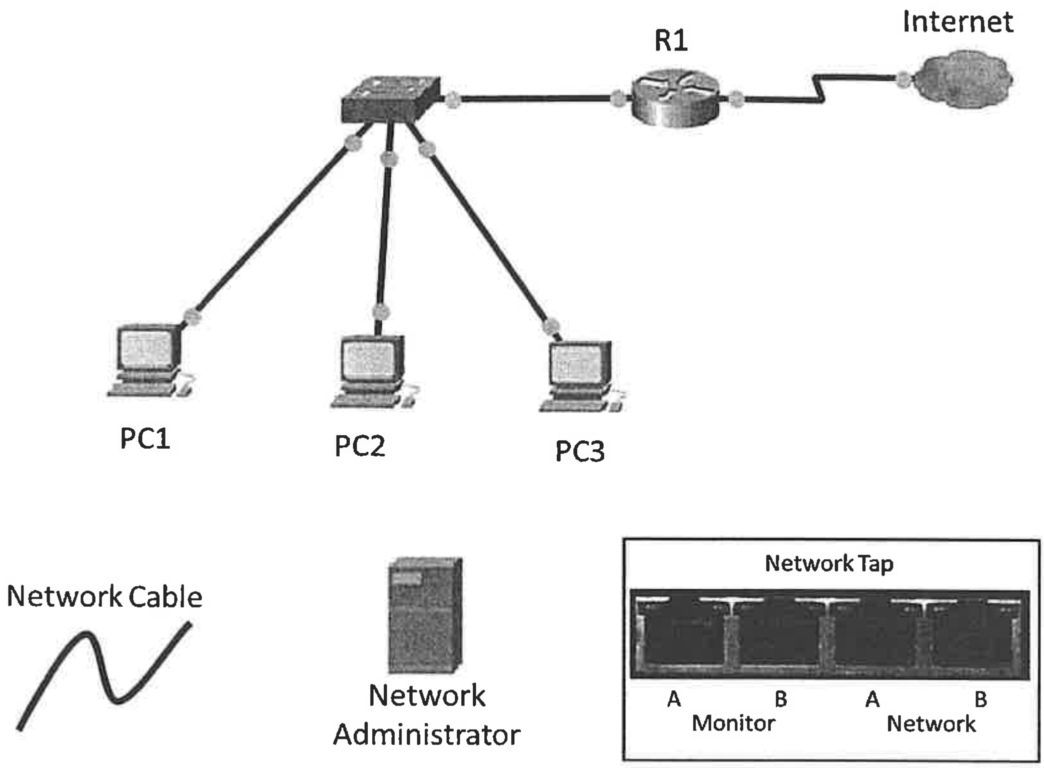


**Figure 4**

[Total : 25 marks]

Q5. (a) Given an IP packet with a total length of 4000 bytes. Assume all the IP packet headers have 4 bytes of options. Answer the following questions:

(i) **Calculate** the size (in bytes) of the initial IP data/payload. (6 marks)

(ii) Suppose the datagram will be routed to an Ethernet network, **determine** the number of fragments that would be created. (2 marks)

(iii) Continue from Q4.(a)(ii), **calculate** the total length for each of the fragments. (6 marks)

(iv) Continue from Q4.(a)(iii), **determine** the fragment offset for each of the fragments. (2 marks)

Q5. (Continued)

(b) Scenario: The receiving TCP announces a window size zero to the sending TCP.

(i) **Describe** how deadlock could occur in this situation. (5 marks)

(ii) **Discuss** how the Persistent Timer is used to prevent deadlock. (2 marks)

(c) **List** TWO (2) services that are provided by the Transport Layer (2 marks)

[Total : 25 marks]